

Simulation of synaptic dynamics in Spiking Neuron

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Abstract

Spiking Neural Networks (SNNs) serve as a bridge between biological realism and computational efficiency. They are considered more biologically realistic than traditional Artificial Neural Networks (ANNs) because they operate using discrete spikes or events, like the way real neurons communicate through action potentials. One of the key advantages of SNNs is their energy efficiency. Because they only communicate when necessary, they can save a lot of energy compared to ANNs, which are constantly processing information. This efficiency makes them attractive for applications where energy consumption is a concern, such as in mobile devices or energy-constrained environments. Another advantage of SNNs is their ability to process information over time, known as temporal processing. This means they can handle input data that changes over time, making them useful for tasks like speech recognition or video processing where the timing of events is important. Additionally, SNNs are highly adaptable to variations in the input data. They can learn and adapt to new information, making them suitable for tasks where the data is not fixed or changes frequently. Overall, SNNs offer a more biologically faithful and energy-efficient approach to neural network computation, making them a valuable tool for understanding brain processes and achieving efficient computation in artificial systems.

Problem Statement

- Replication of the experimental device dynamics with a simple LIF model using STDP.
- Implementation of digit recognition through unsupervised learning using Spike Time Dependent Plasticity (STDP).

Relevance in Neuromorphic Devices

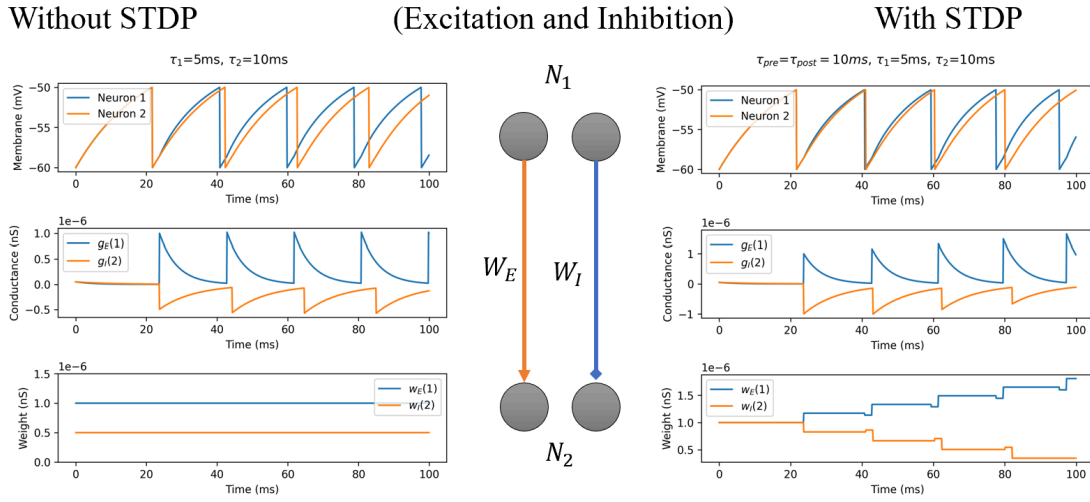
The relevance of modeling synaptic dynamics using Spiking Neural Networks (SNNs) based on Spike Time Dependent Plasticity (STDP) rules lies in its potential application in neuromorphic devices. Through the simulation of biological synapses, these models can facilitate the development of effective and neurally-inspired computing systems. To emulate the workings of the human brain, neuromorphic technologies provide benefits including reduced power consumption, parallel processing, and adaptability to different inputs. By utilizing the effectiveness and resilience of biological systems, STDP-based SNNs in neuromorphic hardware may help us achieve our goal of fully utilizing artificial intelligence while advancing fields like pattern recognition, unsupervised learning, and cognitive computing. Moreover, as the cited paper points out, the incorporation of device-level properties like those of n-channel and p-channel MoS2 transistors may improve the efficacy and realism of these neuromorphic systems, permitting more precise emulation of synaptic responses and aiding in the creation of sophisticated cognitive computing architectures.

Methodology

Implementation of SNN based image classification: For our image classification(digit recognition) project using a Spiking Neural Network (SNN), we used the MNIST database, which contains 60,000 grayscale handwritten digit images sized 28x28 pixels. First, we normalized these images before feeding them into the network. The network has a set of input neurons, each connected to a specific output neuron. We used the NeuronGroup method to implement this connectivity, which required defining a set of differential equations as parameters for each neuron.

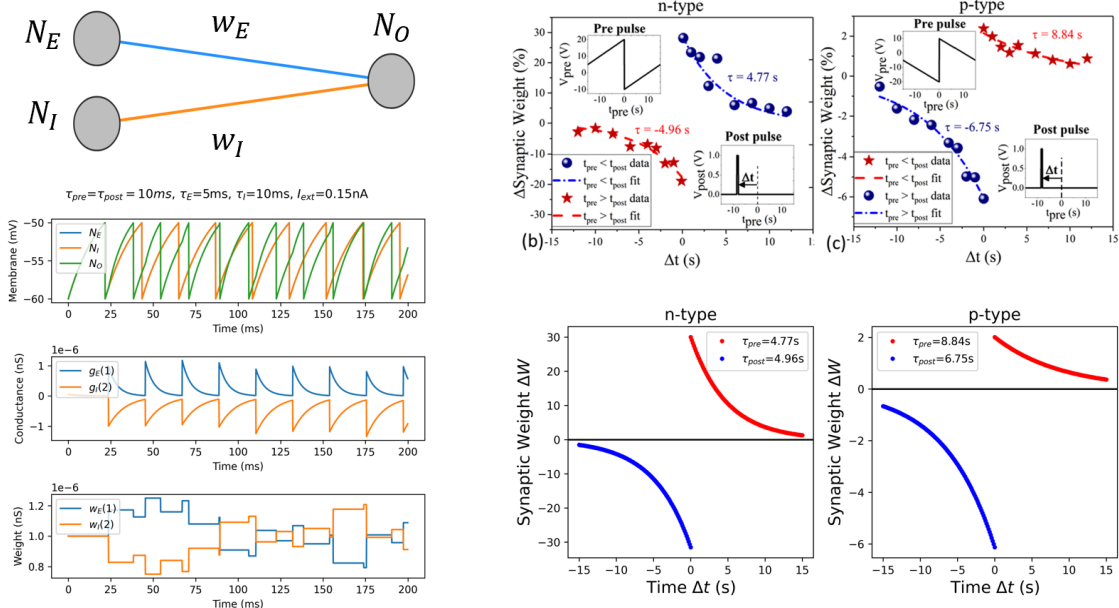
By adjusting the parameters of the model equations, we can change the behavior of the SNN. We then formed synapses between all input neurons and output neurons. These synapses updated their weights based on the timing of pre- and post-neuron spikes. To prevent weights from saturating or becoming very negative, we set upper and lower bounds on the weights and used a small Apre value. During testing, we loaded images from the database, normalized them, and fed them into the network. We used the trained weights for inference. The output of the network was in the form of spikes, and we counted the number of spikes to determine the output class. Overall, this process allowed us to train and test our SNN for image classification tasks using the MNIST dataset. The result has been attached in the Results Section.

Results 1.



Pre and postsynaptic neurons with excitatory and inhibitory synapse without STDP with constant weights and with STDP weights increasing and decreasing for excitation and inhibition neuron pairs.

2.



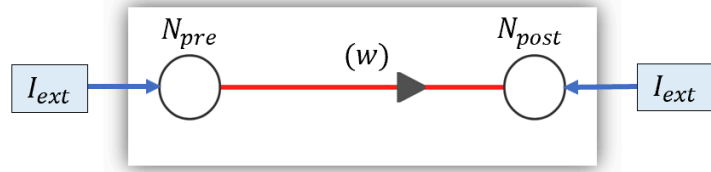
On Left: Neural circuit with excitatory (E) and inhibitory (I) input neurons connected to a single output neuron (O) using STDP with symmetric weights. On right: Replication of experimental data from paper.

3.

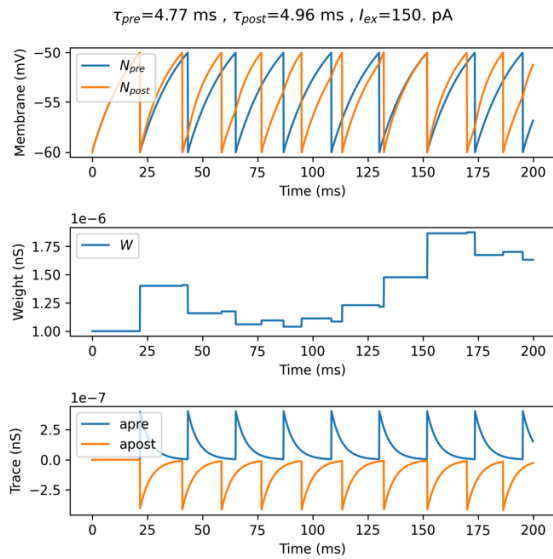
Implementing STDP for n-type and p-type device parameters τ_{pre} and τ_{post} from papers data.

STDP of a Synapse

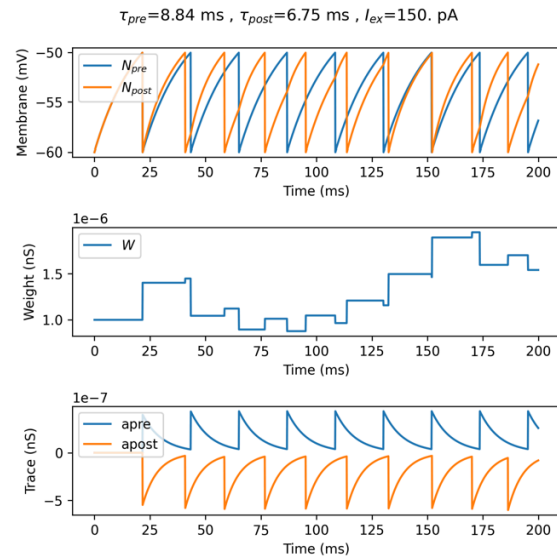
Pre synaptic (N_{pre}) and post synaptic (N_{post}) neuron, weight (w) with external input (I_{ext}).



n-type



p-type



4. SNN Implementation

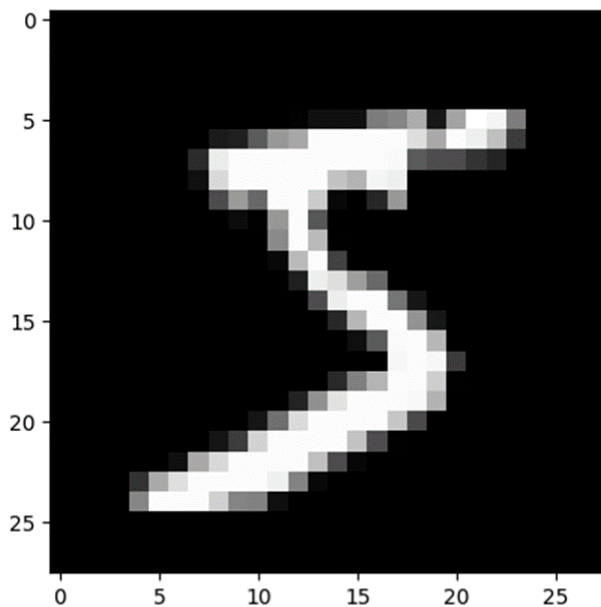


Fig4. Normalized Input Image

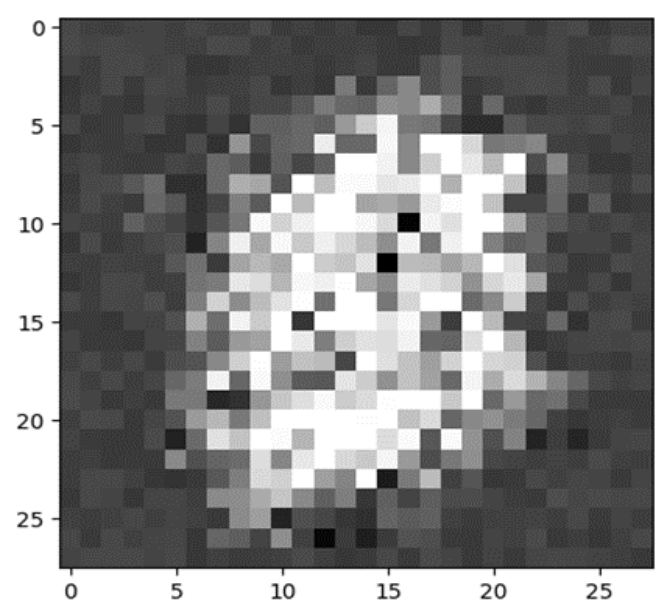


Fig5. Trained Weights post training

Table 1: SNN classification(Recorded output spikes by changing the variabilities)

Parameter	sigma = 0.3, wmax = 1.0, wmin = -1.0, Apre = 0.4*sigma, Apost = -0.2*sigma, vr = 0*volt, vt = 1.5*mV, tau = 10*ms, taupre = 5*ms, taupost = 8*ms, refractoryPeriod=0.1													
Class	0	1	2	3	4	5	6	7	8	9	Sd	u	V	SEM
Output spikes	72	68	77	85	63	74	75	72	79	70	5.78	73.5	33.45	1.8289
Parameter	sigma = 0.1													
Output spikes	47	37	46	56	32	47	48	42	47	48	6.27	45	39.4	1.9849
Parameter	sigma = 0.01													
Output spikes	17	15	23	28	14	22	21	20	22	19	3.91	20.1	15.29	1.2365
Parameter	sigma = 0.01, refractoryPeriod=0.05													
Output spikes	23	14	21	25	15	22	23	19	25	20	3.60	20.7	13.01	1.1406
Parameter	sigma = 0.3, refractoryPeriod=0.05													
Output spikes	77	71	82	91	75	78	79	77	79	72	5.31	78.1	28.29	1.6819
Parameter	sigma = 0.1, refractoryPeriod=0.05													
Output spikes	40	34	41	56	21	47	40	34	44	39	8.68	39.6	75.44	2.7466
Parameter	sigma = 0.1, refractoryPeriod=0.05, tau=20													
Output spikes	53	38	55	59	30	54	56	45	53	48	8.60	49.1	74.09	2.7219
Parameter	sigma = 0.1, refractoryPeriod=0.05, tau=5													
Output spikes	36	31	45	54	31	38	42	34	43	37	6.72	39.1	45.29	2.1281
Parameter	sigma = 0.1, refractoryPeriod=0.05, tau=10, taupost = 8*ms													
Output spikes	49	38	56	67	34	53	50	45	52	47	8.74	49.1	76.49	2.7656
Parameter	sigma = 0.1, refractoryPeriod=0.05, tau=10, taupost = 6*ms													
Output spikes	67	61	63	72	50	65	65	68	73	47	8.11	63.1	65.89	2.5669
Parameter	sigma = 0.1, refractoryPeriod=0.05, tau=10, taupost = 10*ms													
Output spikes	26	9	32	35	24	25	32	29	23	47	9.26	28.2	85.76	2.9284

Conclusion

Simulated a simple leaky integrate and fire model consisting of presynaptic and postsynaptic neurons with excitatory and inhibitory synapse as inputs and single output. Also utilized parameters from an experimental device data to compare STDP based learning rules with weights variations indicating synaptic plasticity. To improve the accuracy of the SNN based network, we need to use optimized parameters for neurons, synapses & weight updation methodology. In our case we have observed that Sigma 0.1, with tau 10, tau post 10ms, tau pre 5ms and refractory period of 0.05 gave us a distinct classification result. We are able to detect digits like 1, 9, 0, 4, 7 distinctly. We can further improve the classification accuracy by fine tuning the other parameters. Also we need to use differential equations & solvers more intuitively as per application. In this project we have designed a neuron, synapse & formed a basic SNN network for handwritten digit classification with required minimum accuracy.

References

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